Type MS · Writing a sigma sum as a formula in n

Answer 1 of 7 · Question 3 part a · 3 marks

2024 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

3(a)	$\left(\sum_{r=1}^{n}(5r-3)\Rightarrow\right)a=2,\quad d=5$	B1
	$\left(\sum_{r=1}^{n}(5r-3)=\right)\frac{n}{2}(2\times2+(n-1)5)$ or $\frac{n}{2}(4+5n-5)$	M1
	or $\frac{n}{2}(2+2+(n-1)5)$ or $\frac{n}{2}(2+5n-3)$	
	$=\frac{n}{2}(5n-1)*$	A1 cso [3]
	ALT (Using standard results)	
	$\left(\sum_{r=1}^{n}(5r-3)=\right)5\sum_{1}^{n}r-3\sum_{1}^{n}1$	B1
	$\left(\sum_{r=1}^{n}(5r-3)=\right)5\left[\frac{n}{2}(n+1)\right]-3n=\frac{5n^2+5n-6n}{2}=\frac{n}{2}(5n-1)*$	M1A1 cso [3]
(b)	$\left(\sum_{31}^{60}(5r-3)=\right)\frac{60}{2}(5\times60-1)-\frac{30}{2}(5\times30-1)=6735$	M1A1 [2]
	ALT	
	$\left(\sum_{31}^{60}(5r-3)=\right)\frac{30}{2}(152+297)=6735$	M1A1 [2]
	or $\left(\sum_{1}^{60}(5r-3)-\sum_{1}^{30}(5r-3)=\right)\frac{60}{2}(2\times2+(60-1)\times5)-\frac{30}{2}(2\times2+(30-1)\times5)=6735$	
(c)	$\frac{n}{2}(5n-1)=3783\Rightarrow5n^2-n-7566=0$	M1
	$\Rightarrow(5n+194)(n-39)=0$	M1
	$\Rightarrow n=39\quad\left[-\frac{194}{5}\right]$	A1 [3]
		Total 8 marks

Type MS · Writing a sigma sum as a formula in n

Answer 1 of 7 · Question 3 part a · 3 marks

2024 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the values of a and d, these may be explicitly stated or implicitly used in a formula.
	M1	Correctly substitutes their values of a and d into the correct summation formula
	A1 cso	For the correct expression, minimum steps as shown, no errors or omissions.
	ALT	
	B1	For writing the given expression as $5\sum_{1}^{n} r - 3\sum_{1}^{n} 1$
	M1	For the sum shown, using standard results for the series.
	A1 cso	For the correct expression, minimum steps as shown, no errors or omissions.
(b)	M1	Uses the given expression with both $n = 60$ and $n = 30$ and subtracts. Indicated by 8970 – 2235. As a concession, allow substitution of $n = 31$ and 60 for the mark. **ALT** Correct substitution of the correct 'first' and 'last' value into the correct summation formula ie $\frac{30}{2}(152+297)$. Allow as a concession $\frac{29}{2}(152+297)$ or correct substitution of both $n = 30$ and $n = 60$ into the correct sum to n terms formula and a subtraction. Indicated by 8970 – 2235. As a concession, allow substitution of $n = 31$ and 60 for the mark.
	A1	For 6735
(c)	M1	For correctly placing the given expression = 3783 and rearranging (allow one error) to get a 3TQ = 0.
	M1	For solving their 3TQ and achieving at least one value. Minimum attempt to solve required (see general guidance). A correct value of 39 can imply this mark
	A1	For 39 only. If $-\frac{194}{5}$ is also given as a solution, withhold this mark.

Type MS · Writing a sigma sum as a formula in n

Answer 1 of 7 · Question 3 part a · 3 marks

2024 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Writing a sigma sum as a formula in n

Answer 2 of 7 · Question 6 part a · 3 marks

2019 JUN · Paper 01 (regional R) · extract 1 of 2 · find part a in this mark scheme extract

Number		
6 (a)	$a = 4\times1-3=1, \quad (d=4)$ $\sum_{r=1}^{n} 4r-3 = \frac{n}{2}\left(2\times1+(n-1)4\right) = n(2n-1)^*$	B1 M1A1 [3]
(b)	$n(2n-1)>1000 \Rightarrow 2n^2-n-1000>0$ $\frac{-(-1)\pm\sqrt{(-1)^2-4\times2\times(-1000)}}{2\times2} \Rightarrow n>22.612... \Rightarrow n=23$	M1 M1A1 [3]
(c)	$3t_{(n+7)}+18 = S_{(n+4)}$ $\Rightarrow 3\left[4(n+7)-3\right]+18=(n+4)\left[2(n+4)-1\right]$ $\Rightarrow 2n^2+3n-65=0$ $2n^2+3n-65=(2n+13)(n-5)=0 \Rightarrow n=5$	 M1 A1 depM1A1 [4]
		Total 10 marks

Type MS · Writing a sigma sum as a formula in n

Answer 2 of 7 · Question 6 part a · 3 marks

2019 JUN · Paper 01 (regional R) · extract 2 of 2 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing a sigma sum as a formula in n

Answer 3 of 7 · Question 5 part a · 3 marks

2020 NOV · Paper 02 · extract 1 of 2 · find part a in this mark scheme extract

5(a)	$\sum_{r=1}^{n}(3r+5)=8+11+14+...+(3n+5)=\frac{1}{2}n(8+3n+5)=\frac{1}{2}n(3n+13)$ *	M1M1A1cso (3)
ALT	$\sum_{r=1}^{n}(3r+5)=\sum_{1}^{n}3r+5n=\frac{n}{2}(3+3n)+5n=\frac{n}{2}(13+3n)$ *	M1M1A1cso
(b)	$\sum_{r=35}^{50}(3r+5)=\frac{50}{2}(13+150)-\frac{34}{2}(13+102)$	M1
	$=2120$	A1 (2)
(c)	$\frac{n}{2}(13+3n)=385$	
	$3n^2+13n-770=0$	M1
	$(3n+55)(n-14)=0 \quad n=14$	M1A1 (3)
		[8]

(a)	
M1	Evaluate **either** first and last terms **or** first and common difference
M1	Use either sum formula. Can be shown explicitly or implied by a correct, full substitution of n and their a and their d or l
A1cso	Reach the **given** result with no errors in the working. Must be the complete result, not just the RHS or there must be a conclusion eg "shown"
ALT	
M1	Split the $(3r + 5)$ into 2 parts and deal with the 5 correctly
M1	Sum $(3r)$ either by using a summation formula or by using the standard result
A1	Reach the **given** result with no errors in the working
(b)	
M1	Express the required sum as the difference of 2 sums. Second sum must have 34 terms. Use the result given in (a). **Using a standard formula** with first term and either last term or common difference scores 0/2 as question states "hence". Calculator solutions (without showing the difference of the 2 sums first) score M0
A1	Correct answer.
(c)	
M1	Use the result in (a) or some other valid method to form a 3 term quadratic in n
M1	Solve their 3TQ by any valid means. Must reach n = ... Negative value need not be seen
A1	Correct answer. n = 14 and no other Correct quadratic followed by correct answer scores 3/3 Incorrect quadratic solved by calculator M0A0

Type MS · Writing a sigma sum as a formula in n

Answer 3 of 7 · Question 5 part a · 3 marks

2020 NOV · Paper 02 · extract 2 of 2 · find part a in this mark scheme extract

Question Number	Scheme	Marks
	$\sum_{r=3}^{5}$	

Type MS · Writing a sigma sum as a formula in n

Answer 4 of 7 · Question 1 part a · 3 marks

2023 JUN · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks
1(a)	$d=3$ and $a=5$	B1
	$\sum_{r=1}^{n}(3r+2)=\frac{n}{2}\left(2\times5+(n-1)3\right)=\frac{n}{2}(3n+7)$*	M1A1 cso [3]
	ALT	
	$\sum_{r=1}^{n}(3r+2)=3\sum_{r=1}^{n}r+2\sum_{r=1}^{n}1$	[B1
	$\sum_{r=1}^{n}(3r+2)=\frac{3}{2}n(n+1)+2n=\frac{3n^2+3n}{2}+\frac{4n}{2}=\frac{3n^2+7n}{2}=\frac{n}{2}(3n+7)$	M1A1]
(b)	$\sum_{r=10}^{40}(3r+2)=\frac{40}{2}(3\times40+7)-\frac{9}{2}(3\times9+7)=[2540-153]=2387$	M1A1 [2]
		Total 5 marks

Note: Part (b) can be answered simply by entering the expression in the question into a permissible calculator. A solution seen with no working is M0.

Type MS · Writing a sigma sum as a formula in n

Answer 4 of 7 · Question 1 part a · 3 marks

2023 JUN · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

Part	Mark	Notes
(a)	B1	For the correct value of a and d
	M1	For using the correct summation formula with their values of a and d
	A1	For achieving $\sum_{r=1}^{n}(3r+2)=\frac{n}{2}(3n+7)$ with no errors
	ALT 1 – Uses first plus last formula	
	B1	For the correct value of $a = 5$ and the final value $l = 3n + 2$
	M1	For the correct use of the first plus last summation formula $\sum_{r=1}^{n}(3r+2)=\frac{n}{2}(5+[3n+2])=\frac{n}{2}(3n+7)$
	A1	For achieving $\sum_{r=1}^{n}(3r+2)=\frac{n}{2}(3n+7)$ with no errors
	ALT 2 – Uses standard results	
	B1	For stating $\sum_{r=1}^{n}(3r+2)=3\sum_{r=1}^{n}r+2\sum_{r=1}^{n}1$
	M1	For using the standard expressions $\sum_{r=1}^{n}(3r+2)=\frac{3}{2}n(n+1)+2n$
	A1	For simplifying to the required form. $\sum_{r=1}^{n}(3r+2)=\frac{3n^2+3n}{2}+\frac{4n}{2}=\frac{3n^2+7n}{2}=\frac{n}{2}(3n+7)$ *
(b)	M1	Uses the given formula with $n = 40$ and with $n = 9$ (allow $n = 10$ for this mark) [Note: use of $n = 10$ will give a value of 2355] Allow alternative correct methods
	A1	For the correct sum of 2387
	ALT 1 – Uses first + last summation formula	
	M1	$U_{10}=32,\quad U_{40}=122,\quad n=31$ (allow 30 for this mark) $\sum_{r=10}^{40}(3r+2)=\frac{31}{2}(32+122)=...$ [Note: Use of $n = 30$ will give a value of 2310]
	A1	For the correct sum of 2387
	ALT 2 – Finds new first term and uses the summation formula	
	M1	$U_{10}=32\quad n=31$ (allow 30 for this mark) $\sum_{r=10}^{40}(3r+2)=\frac{31}{2}(2\times32+(31-1)3)=...$ [Note: Use of $n = 30$ will give a value of 2265]
	A1	For the correct sum of 2387

Type MS · Writing a sigma sum as a formula in n

Answer 4 of 7 · Question 1 part a · 3 marks

2023 JUN · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question	Scheme	Marks

Type MS · Writing a sigma sum as a formula in n

Answer 5 of 7 · Question 9 part a · 3 marks

2021 NOV · Paper 01 · extract 1 of 3 · find part a in this mark scheme extract

Number		
9 (a)	$a = 5\times1-1=4$ $d=5$ $S_n=\frac{n}{2}\left(2\times4+[n-1]5\right)=\frac{n}{2}(3+5n)*$ cso **ALT** $a=4$ $l=5n-1$ $S_n=\frac{n}{2}(a+l)=\frac{n}{2}(4+5n-1)=\frac{n}{2}(3+5n)*$ cso	B1 M1A1 [3] [B1 M1A1]
(b)	$\sum_{r=10}^{20}(5r-1)=\sum_{r=1}^{20}(5r-1)-\sum_{r=1}^{9}(5r-1)$ $\sum_{r=10}^{20}(5r-1)=\frac{20}{2}(3+5\times20)-\frac{9}{2}(3+5\times9)$ $=1030-216=814$ **ALT** $a=4+9\times5=49,\ l=4+20\times5=99,\ n=20-10+1=11$ $\sum_{r=10}^{20}(5r-1)=\frac{11}{2}(49+99)=814$ **ALT** $a=5\times10-1=49$ $d=5$ $n=11$ $S_n=\frac{11}{2}(2\times49+(11-1)\times5)=814$	B1 M1 A1 [3] [B1 M1A1] [B1 M1A1]
(c)	$\frac{n}{2}(3+5n)=12(4+5n)+52\Rightarrow5n^2-117n-200=0$ $\Rightarrow(n-25)(5n+8)=0\Rightarrow n=25,\ \left(n\neq-\frac{8}{5}\right)$	M1M1A1 M1A1 [5]
		Total 11 marks

Type MS · Writing a sigma sum as a formula in n

Answer 5 of 7 · Question 9 part a · 3 marks

2021 NOV · Paper 01 · extract 2 of 3 · find part a in this mark scheme extract

		May be implied by correct values seen in summation formula.
	M1	Uses a **correct** form of the summation formula for an arithmetic series with their a and their l provided their a and their l are stated. $S_n = \frac{n}{2}(a+l) = \frac{n}{2}('4' + '5n - 1')$
	A1 cso	For obtaining the given answer in full with no errors. $\sum_{r=1}^{n}(5r-1) = \frac{n}{2}(3+5n)$
Note: If standard summation results are correctly used award B1M1A1, if not fully correct send to review.		
(b)	**B1**	For correctly giving the required summation as the difference between two summations starting at $r = 1$. $\sum_{r=10}^{20}(5r-1) = \sum_{r=1}^{20}(5r-1) - \sum_{r=1}^{9}(5r-1)$
	M1	For substitution of $n = 20$ and $n = '9'$ into the result from part (a) and subtracting. $\frac{20}{2}(3+5\times 20) - \frac{'9'}{2}(3+5\times' 9')$ Allow for use of 9 or 10.
	A1	For the correct summation 814
Alternative method		
	B1	For finding the first term, last term and number of terms for the arithmetic sequence. $a = 49, l = 99, n = 11$
	M1	Uses a **correct** summation formula for an arithmetic series with their a, their $l$and their n provided these are stated. $\frac{'11'}{2}('49'+'99')$
	A1	For the correct summation 814
Alternative method – considering this as a series starting at the 10th term of the original series		
	B1	For finding the first term, common difference and number of terms. $a = 5\times 10 - 1 = 49$ $d = 5$ $n = 11$
	M1	Uses a **correct** form of the summation formula for an arithmetic series with their a, *their* d, and *their* n provided their a and their d and their n are stated and their $n \neq 20$ $S_n = \frac{'11'}{2}(2\times '49' + ('11'-1)\times '5')$
	A1	For the correct summation 814
(c)	**M1**	Uses $5r-1$ with $n+1$ to find an expression for u_{n+1} in terms of n. $5r-1 = 5(n+1)-1 = 5n+4$
	M1	Forms a correct equation for n using the result given in part (a) and their expression for $5r-1$ in terms of n. $\frac{n}{2}(3+5n) = 12('4+5n') + 52$
	A1	Obtains a correct 3TQ $5n^2 - 117n - 200 = 0$ oe
	M1	For an attempt to solve their 3TQ. See General Guidance for the definition of an attempt. $(n-25)(5n+8) = 0 \Rightarrow n = 25, \left(n = -\frac{8}{5}\right)$
	A1	For correct value: $n = 25$ If $n = -\frac{8}{5}$ is seen it must be rejected.

Type MS · Writing a sigma sum as a formula in n

Answer 5 of 7 · Question 9 part a · 3 marks

2021 NOV · Paper 01 · extract 3 of 3 · find part a in this mark scheme extract

Question number	Scheme	Marks

Type MS · Writing a sigma sum as a formula in n

Answer 6 of 7 · Question 6 part b · 6 marks

2022 JAN · Paper 02 · extract 1 of 3 · find part b in this mark scheme extract

Question	Scheme	Mark
6(a)	$S_{20} = 20(3+2\times 20) = 860$	M1A1 [2]
(b)	$S_1 = 3\times 1+2\times 1^2 = 5$ $S_2 = 3\times 2+2\times 2^2 = 14$ $14 = 5+U_2 \Rightarrow U_2 = 9$ $d = 9-5 = 4$ $U_n = '5'+(n-1)'4'$ $S_n = \sum_{r=1}^{n}(4r+1) \Rightarrow A = 4, \quad B = 1$	B1 M1 M1 A1 M1 A1 [6]
(c)	$T_n = \frac{n}{2}(2\times 7+(n-1)4) = \left[\frac{n}{2}(10+4n)\right] \Rightarrow T_n = \frac{n}{2}(10+4n)$ or $T_n = n(5+2n)$ $n(5+2n) = 3n+2n^2+252, \quad 5n = 2n+252$ $5n = 3n+252 \Rightarrow n = 126$	M1 M1,dM1 ddM1A1 [5]
	Total	**13 marks**

Type MS · Writing a sigma sum as a formula in n

Answer 6 of 7 · Question 6 part b · 6 marks

2022 JAN · Paper 02 · extract 2 of 3 · find part b in this mark scheme extract

Part	Mark	Notes
6(a)	M1	For substituting 20 into the given $S_{20} = 20(3+2\times 20) = 860$
	A1	For $S_{20} = 860$ Sight of 860 with no working scores M1A1
(b)	B1	For finding the first term $S_1 = 3\times 1+2\times 1^2 = 5$ or $a = 5$ Award this mark even if it is not clear that they understand that S_1 is the first term.
	M1	For a complete method to find the second term $S_2 = 3\times 2+2\times 2^2 = 14$ $14 = 5+U_2 \Rightarrow U_2 = 9$
	M1	For finding the common difference, they must reach a value for this mark. $d = '9'-'5' = (4)$
	A1	For $d = 4$
	M1	For either $A = 4$ or $B = 1$ Accept embedded values.
	A1	For both $A = 4$ and $B = 1$ Accept embedded values.
	For the final correct answer seen without any or minimal working, award full marks in part (b)	
(c)	M1	For an expression for T_n using the given values $T_n = \frac{n}{2}(2\times 7+(n-1)4) = \left[\frac{n}{2}(10+4n)\right]$
	M1	For equating their expression for T_n in $T_n = n(3+2n)+252$ The correct expression for S_n must be used here. $\Rightarrow \frac{n}{2}(10+4n) = n(3+2n)+252 \Rightarrow (5n = 3n+252)$ **This is an A mark in Epen**
	dM1	For forming a linear equation. e.g. $5n = 3n+252$ o.e. This mark is dependent on the previous M mark.
	ddM1	For solving **their** linear equation. $5n = 3n+252 \Rightarrow n = (126)$ This mark is dependent on both previous M marks.
	A1	For $n = 126$

Type MS · Writing a sigma sum as a formula in n

Answer 6 of 7 · Question 6 part b · 6 marks

2022 JAN · Paper 02 · extract 3 of 3 · find part b in this mark scheme extract

Question	Scheme	Mark

Type MS · Writing a sigma sum as a formula in n

Answer 7 of 7 · Question 4 part a · 3 marks

2025 NOV · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

4(a)	$\sum_{r=1}^{n}(4r+A)=\frac{n}{2}\left(2(4+A)+(n-1)4\right)=\frac{n}{2}(4+2A+4n)$ Then $\frac{n}{2}(4+2A+4n)=\frac{n}{2}(4n-6)$ $4+2A=-6\Rightarrow A=-5$*	M1 M1 A1 cso [3]
(b) (i)	$\sum_{21}^{60}(4r-5)=\frac{60}{2}(4\times60-6)-\frac{20}{2}(4\times20-6)=[7020-740]=6280$ **ALT 1** $U_{21}=4\times21-5=79 \quad U_{60}=4\times60-5=235$ $\sum_{21}^{60}(4r-5)=\frac{40}{2}(79+235)=6280$ **ALT 2** $S_{40}=\frac{40}{2}\left(2(79)+(40-1)\times4\right)$ $=20\times(158+156)=6280$	M1A1 [2] [M1A1] [M1A1]
(ii)	$\sum_{r=1}^{n}(4r-5)<3418\Rightarrow\frac{n}{2}(4n-6)<3418\Rightarrow4n^2-6n-6836<0$ $\Rightarrow4n^2-6n-6836<0\Rightarrow n<42.096....$ $\Rightarrow n=42$	M1 M1 A1 [3]
		Total 8 marks

Type MS · Writing a sigma sum as a formula in n

Answer 7 of 7 · Question 4 part a · 3 marks

2025 NOV · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question	Scheme	Marks